Toward Realistic Evaluation of GNSS Anti-Spoofing: A Spoofing Dataset from Urban Experiments

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BIOGRAPHY

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ABSTRACT

Global navigation satellite systems (GNSS) spoofing poses a serious threat to satellite-based navigation systems, especially when synchronous spoofing is used to smoothly take over an already locked receiver without causing obvious signal discontinuities. Although many anti-spoofing methods have been developed, their validation still relies on a limited number of public datasets or simplified simulation scenarios. Existing datasets provide valuable references, but they do not fully capture the degraded signal reception conditions commonly encountered in dense urban environments. Environmental effects can produce receiver-level signatures similar to spoofing-induced anomalies and may also force spoofing detectors to adopt more conservative thresholds to avoid false alarms. In particular, small-position-error spoofing attacks may be hidden within the normal urban positioning error floor. To address this gap, this paper presents HK-UrbanSpoof, an open GNSS spoofing dataset collected in urban environments. The dataset includes four experimental environments, ranging from open-sky and coastal suburban to dense urban scenarios, and contains 17 synchronous spoofing tests with different receiver dynamics, spoofing power advantages, and spoofed position displacements. Time-domain labels for multipath, signal blockage, and non-line-of-sight (NLOS) reception are provided by ray-tracing analysis based on three-dimensional building models. Raw intermediate-frequency recordings and receiver true trajectories are also released to support anti-spoofing research. HK-UrbanSpoof provides a challenging benchmark for evaluating the robustness and generalizability of GNSS anti-interference techniques under complex urban scenarios.

I. INTRODUCTION

Global navigation satellite systems (GNSS) provide the primary source of absolute position, velocity, and time information for a wide range of civil, commercial, and safety-critical applications. The increasing dependence on GNSS also exposes these applications to intentional radio frequency (RF) interference. In recent years, an increasing number of incidents have been reported involving intentional GNSS interference affecting civilian users, influencing map applications, taxi sharing, and food delivery platforms (Newsroom, 2024; Stojkovski, 2025). GNSS signals are intrinsically vulnerable to spoofing because their signal structures are publicly known and their received power at the antenna is extremely weak. As a result, counterfeit GNSS-like signals can be generated using inexpensive software-defined radios and commercial-off-the-shelf (COTS) hardware (Humphreys et al., 2008; Psiaki and Humphreys, 2021). In a self-consistent spoofing attack, a transmitter broadcasts counterfeit satellite signals corresponding to a false position, causing receivers within the affected area to estimate an erroneous location (Psiaki and Humphreys, 2016). More sophisticated attacks attempt to gradually align the counterfeit signals with the authentic signals and smoothly take over the receiver tracking loops. This type of attack, commonly referred to as synchronous or seamless spoofing, is especially threatening because it may avoid abrupt power changes, loss of lock, or obvious tracking anomalies during the initial pull-off stage.

Despite substantial progress in spoofing detection and mitigation techniques, a fundamental obstacle remains: the lack of open, realistic, and well-documented GNSS spoofing datasets. This limitation is mainly caused by the difficulty of constructing

controllable spoofing testbeds, generating sufficiently synchronized counterfeit signals, and obtaining reliable ground truth under real propagation conditions. Consequently, many existing studies rely on a small number of public datasets for algorithm validation. The Texas Spoofing Test Battery (TEXBAT) is one of the earliest and most widely used public datasets for GNSS spoofing research (Humphreys et al., 2012). The Oak Ridge Spoofing and Interference Test Battery (OAKBAT) extends this line of work by providing spoofing scenarios for the Galileo E1 signal (Albright et al., 2020). However, both the authentic and spoofing signals in OAKBAT are generated by simulators, and the dataset is not currently available as a fully open-access resource until July 2026. The FGI-SpoofRepo further provides open raw intermediate frequency (IF) recordings for multiple frequencies and constellations, including GPS L1, Galileo E1, GPS L5, and Galileo E5a, and covers synchronous, asynchronous, and meaconing scenarios (Islam et al., 2024). Most recently, the Tuni2025 GNSS dataset has focused on asynchronous spoofing attacks on GPS L1 and Galileo E1, with spoofed pseudorandom codes (PRNs) are intentionally different from visible live-sky PRNs (Rahman et al., 2025).

However, several gaps still limit the use of existing datasets for evaluating anti-spoofing techniques in practical urban environments. First, existing studies heavily rely on a small number of public datasets or simplified simulation settings for performance validation. Existing anti-spoofing methods span different processing levels, including external-hardware-aided approaches, receiver signal-processing methods, measurement-domain methods, and machine-learning-based detectors. However, our statistical examination of some existing studies shows that approximately 50% of the methods are validated exclusively on TEXBAT (Chen et al., 2025, 2026; Fang and Xu, 2025; Fang et al., 2023; Iqbal et al., 2024; Li et al., 2025b; Rothmaier et al., 2021; Shang et al., 2022a,b, 2023; Sun et al., 2021; Wang et al., 2017, 2026, 2023), while another 11% rely on both TEXBAT and OAKBAT (Chen et al., 2022; Li et al., 2025a; Mao et al., 2025). Approximately 14% rely on simulated spoofing scenarios, where spoofing effects are approximated by imposing pre-designed false trajectories (Bai et al., 2024; Ceccato et al., 2021; Pardhasaradhi et al., 2022; Qian et al., 2026). Although some studies have conducted dedicated spoofing experiments, the corresponding datasets are generally not publicly available (Broumandan et al., 2015; Gross et al., 2019; Shafiee et al., 2018; Zarrinnegar et al., 2026). Other recent studies have been validated using Jammertest, an authorized over-the-air GNSS interference field campaign in Norway (Liu and Papadimitratos, 2025; Spanghero et al., 2025). These datasets, simulations, and field campaigns have provided valuable benchmarks for the development and comparison of GNSS anti-spoofing techniques. To more comprehensively assess the robustness and generalizability of anti-spoofing methods, additional open datasets collected from diverse environments, receiver platforms, user dynamics, and attack strategies are still needed.

Second, existing spoofing datasets, including TEXBAT, OAKBAT, FGI-SpoofRepo, and Tuni2025-GNSS, are mainly collected under controlled laboratory conditions or relatively benign field environments. As summarized in Table 1, TEXBAT, OAKBAT, and FGI-SpoofRepo do not define controlled multipath scenarios. Tuni2025-GNSS considers limited simulated multipath in the spoofing signals, with range delays of 200–500 m. For GPS L1 C/A, this corresponds to approximately 0.68–1.7 code chips, meaning that many simulated reflections may fall outside the effective tracking-discriminator region and therefore may not represent the short-delay multipath effects that typically bias receiver tracking in dense urban areas. Although these datasets provide important references for GNSS spoofing research, they do not fully capture the degraded reception conditions commonly encountered in realistic urban navigation. In dense urban environments, the victim receiver can be simultaneously affected by multipath, non-line-of-sight (NLOS) reception, diffraction, intermittent blockage, receiver dynamics, low carrier-to-noise-density ratio (C/N_0), and occasional tracking instability. These effects can distort the ACF, bias pseudorange and Doppler measurements, fluctuate C/N_0, and destabilize tracking-loop outputs, producing receiver-level signatures that overlap with spoofing-induced anomalies. Consequently, methods that perform well under clean spoofing conditions may suffer from high false alarm rates, missed detections, or unstable parameter estimation when spoofing coexists with severe urban propagation effects.

Third, GNSS spoofing detectors based on position-domain consistency checks usually compare the GNSS-derived position or trajectory with auxiliary information from IMUs, odometers, network-based positioning, or other independent sensors. These methods assume that spoofing will introduce a measurable inconsistency between the GNSS solution and the auxiliary motion. However, urban propagation effects and thermal noise can naturally enlarge GNSS positioning errors, widening the nominal error distribution and requiring higher detection thresholds to avoid false alarms (Clements et al., 2022). Consequently, if an attacker intentionally maintains a small target position error, the spoofing-induced deviation may be hidden within the urban error floor and become difficult to distinguish from ordinary environmental effects. This is particularly critical for subtle attacks that mimic urban disturbances or primarily target timing errors. Therefore, real-world urban spoofing datasets with small-position-error attacks are needed, where spoofing, multipath, blockage, and receiver dynamics coexist.

Finally, synchronous spoofing is especially challenging because it attempts to take over an already locked receiver without causing obvious signal discontinuities. However, some existing spoofing recordings still exhibit residual asynchrony between authentic and spoofing signals. For example, although FGI-SpoofRepo is sufficiently synchronized to avoid loss of lock, disturbances have been reported in the prompt correlator outputs, carrier tracking loop outputs, and Doppler observables. This may indicate that the receiver is transferred relatively quickly from authentic-signal tracking to the spoofed trajectory, rather than being gradually dragged over a longer interval. Such residual inconsistencies can introduce abrupt measurement changes and thereby reduce the difficulty of detection.

Table 1: Comparison of GNSS spoofing datasets.

Characteristic	TEXBAT	OAKBAT	FGI-SpoofRepo	Tuni2025 GNSS	HK-UrbanSpoof
Measurement year	2012	2020	2024	2025	2026
Location	Texas, USA	Tennessee, USA	Espoo, Finland	Tampere, Finland	Hong Kong, China
Spoofed signals	GPS L1	GPS L1; Galileo E1	GPS L1; Galileo E1; GPS L5; Galileo E5a	GPS L1; Galileo E1	GPS L1
No. of spoofing scenarios	8	12	8	13	17
Live-sky GNSS signal	Yes	Simulated	Yes	Yes	Yes
Clean signal baseline	Yes	Yes	No	Yes	Yes
Receiver motion	Static; Dynamic vehicle	Static; Simulated dynamic route	Static	Static	Static; Dynamic pedestrian; Dynamic vehicle
Attack type	Synchronous spoofing	Synchronous spoofing	Synchronous spoofing; Asynchronous spoofing; Meaconing	Asynchronous spoofing	Synchronous spoofing
Code-carrier consistency	Yes; No	No	Yes	Yes	Yes
Multipath included	No	No	No	Yes, simulated	Yes, real urban
Reference	Humphreys et al. (2012)	Albright et al. (2020)	Islam et al. (2024)	Rahman et al. (2025)	This article

To address these gaps, this work introduces an open-source GNSS spoofing dataset collected in dense urban environments (HK-UrbanSpoof), where synchronous spoofing attacks coexist with authentic GNSS signals, severe multipath propagation, intermittent signal blockage, and NLOS reception. Beyond spoofing detection, the dataset is intended to support interference classification, spoofing- and multipath-related parameter estimation, and the development of measurement- and navigation-domain correction methods for resilient GNSS positioning. The main contributions are summarized as follows:

1) We develop an urban GNSS spoofing data collection system, including the authentic GNSS signal reception platform, spoofing signal generation platform, and post-processing analysis workflow. The dataset covers diverse spoofing scenarios categorized by signal reception condition, victim receiver dynamics, spoofing power advantage, and spoofed position displacement.
2) We provide ground-truth labels for reliable anti-spoofing evaluation. The dataset includes time-domain labels for spoofing signal presence, multipath interference, signal blockage, and NLOS reception. Ray-tracing based on three-dimensional (3D) building models is used to identify environmental effect labels.
3) A high-grade integrated navigation system provides centimeter-level reference trajectories for navigation-domain validation. For each scenario, time-tagged position, velocity, attitude information of the antenna, raw IF data of the clean authentic signal, and the spoofed signal are provided, enabling signal-level analysis, parameter estimation, and mitigation-oriented validation

II. SPOOFING ATTACK DESIGN AND DATA-COLLECTION SETUP

1. Synchronous Spoofing Attack Model

The spoofing scenarios in this dataset are designed as warm-start synchronous spoofing attacks. Before injection, the victim receiver is already locked onto the authentic GNSS signals and outputs stable navigation solutions for at least 36 s. The spoofing signal is then injected with negligible power and gradually amplified until it overshadows the authentic signal. After the initial alignment, the spoofer enters the takeover, or pull-off, stage, where the spoofing code phase and Doppler frequency are smoothly shifted according to the predefined false trajectory. This attack model requires temporal alignment between the spoofing and authentic GNSS signals at injection, which in practice relies on prior knowledge of the approximate victim position and careful signal synchronization. During pull-off, the spoofing Doppler is adjusted consistently with the spoofed range-rate evolution:

$$f_D = -\frac{\dot{\rho}}{\lambda}, \tag{1}$$

where f_D is the spoofing Doppler frequency, $\dot{\rho}$ is the pseudorange rate, and λ is the carrier wavelength. This strategy follows the *frequency-unlocked attack mode* used in TEXBAT, where the spoofing Doppler evolves with the spoofed code-phase change rate rather than being locked to the authentic carrier phase. The frequency-locked mode is not considered because it requires precise authentic carrier-phase knowledge and may introduce code-carrier inconsistencies that can be exploited by measurement-domain spoofing detectors.

2. Spoofing Signal Generation and Transmission

Fig. 1 illustrates the spoofing signal generation and transmission workflow. The spoofing signal generator includes two main modules: a trajectory/timing controller and a GNSS baseband signal simulator. The controller defines the spoofed trajectory and derives the timing parameters required for alignment with the authentic GNSS signal. Broadcast ephemeris data are obtained from NASA's Crustal Dynamics Data Information System (CDDIS) in RINEX format and used by the simulator to synthesize digitized baseband spoofing samples. The simulator is developed based on GPS-SDR-SIM, an open-source GNSS signal simulator for generating GPS baseband I/Q samples (Ebinuma, accessed: 2025). The generated samples are then upconverted to RF and transmitted using COTS software-defined radio hardware, such as the USRP.

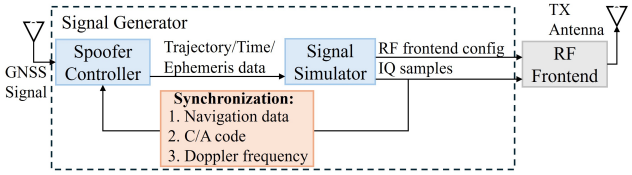


Figure 1: Spoofing signal generation and transmission workflow.

The spoofing experiments are conducted in a shielded anechoic chamber providing more than 100 dB attenuation. Two RF channels are used: one replays the authentic GNSS signal, while the other transmits the generated spoofing signal. To avoid ethical, safety, and legal risks associated with over-the-air GNSS spoofing, authentic GNSS signals are first recorded in outdoor urban environments and then replayed in the laboratory together with the spoofing signals. This replay-based strategy enables repeatable spoofing experiments while preserving realistic reception effects, including multipath, NLOS reception, signal blockage, and receiver dynamics. Fig. 2(a) and (b) show the overall test flow and laboratory implementation. Both authentic and spoofing signal streams are transmitted using USRP N210 devices. The victim receiver antenna receives the superimposed

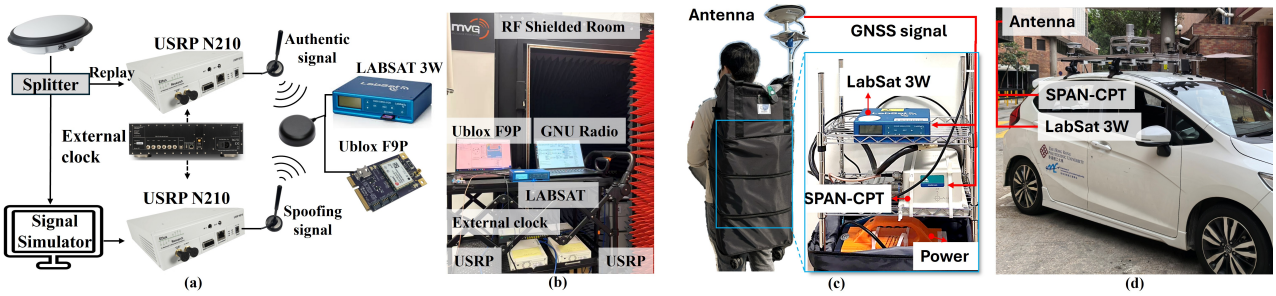


Figure 2: Laboratory spoofing transmission testbed: (a) overall experimental flow and (b) implementation inside the shielded anechoic chamber. Authentic GNSS signal recording platforms for urban data collection: (c) static and pedestrian scenarios and (d) vehicle scenarios.

signal, which is split into two branches. One branch is connected to a LabSat 3 Wideband (3W) recorder for raw IF data collection, while the other is connected to a commercial u-blox F9P receiver for real-time monitoring of the victim receiver navigation performance.

The dataset covers representative urban spoofing scenarios. Fig. 2(c) and (d) show the authentic GNSS signal recording platforms for the static/pedestrian and vehicle scenarios. A NovAtel GPS-703-GGG antenna is mounted on the platform and connected to a signal splitter. One output branch is connected to a LabSat 3W for raw GNSS signal recording, and the other is connected to a NovAtel SPAN-CPT integrated navigation system, which provides centimeter-level reference positions for ground-truth trajectory generation. The same architecture is used for the vehicle platform. Table 2 summarizes the configuration of the public I/Q data.

Table 2: Configuration of the IF data

Parameter	Center frequency	Intermediate frequency	Sampling rate	Data type	Bandwidth
Value	1580 MHz	−4.58 MHz	58 MHz	8 bit (I/Q)	56 MHz

3. Signal Synchronization

Successful injection into the tracking loop of an already locked receiver requires close temporal and frequency alignment between the authentic GNSS signal and the generated spoofing signal. In this testbed, synchronization is handled at two levels: sample-level synchronization in the spoofing signal controller and RF-front-end synchronization between the two transmission channels. At the sample level, the spoofing signal controller estimates the synchronization parameters of the recorded authentic GNSS signal and uses them to generate the spoofing I/Q samples. A short segment of the authentic signal is tracked and decoded to obtain the navigation data bits, C/A-code phase, and Doppler frequency for each satellite channel. Since the experiments are conducted in a post-processing laboratory replay environment rather than in real time, the receiver clock bias during authentic-signal recording must be estimated and reproduced during spoofing signal generation. The spoofing signal sampling rate is adjusted to compensate for residual clock drift and code-phase mismatch. A higher sampling rate improves the temporal resolution of this adjustment. For GPS L1 C/A, one code chip corresponds to approximately 977.5 nanoseconds (ns), or about 293 m in range. Based on our experimental experience, keeping the initial spoofing-authentic signal alignment within approximately 100 ns effectively reduces tracking abnormalities during injection. At the RF-front-end level, the two USRP N210 transmitters are synchronized using an Ettus OctoClock-G CDA-2990 timing distribution unit. The OctoClock-G distributes a common 10 MHz frequency reference and 1-pulse per second (PPS) timing signal from its internal GPS-disciplined oven-controlled crystal oscillator (GPSDO/OCXO) to both USRPs, ensuring a common time and frequency base. Nevertheless, spoofing takeover failures were still observed in repeated trials, possibly due to residual hardware-induced instability, including temperature-dependent GPSDO drift, imperfect RF-chain calibration, cable-delay mismatch, and small residual timing or frequency offsets between the two channels (Sathaye et al., 2022). Therefore, repeated calibration and spoofing trials were conducted, and only recordings with stable injection and successful takeover were included in the dataset.

III. CLEAN AND AUTHENTIC SCENARIOS

The dataset is built on four clean authentic-signal scenarios, which are used to construct 17 spoofing tests with different attack strategies. These clean recordings include three real-world scenarios and one simulation-based case, covering different receiver dynamics, propagation environments, and reception conditions, from open-sky to suburban and dense urban canyon environments. The recording environments, victim receiver trajectories, and sky plots are shown in Fig. 3. Scenario 1 is a



Figure 3: The test environment, victim receiver trajectory, and sky plot for (a) scenario 1, (b) scenario 2, (c) scenario 3, and (d) scenario 4. Green satellites indicate line-of-sight (LOS) reception, while orange satellites indicate multipath-affected signals identified by ray-tracing analysis.

static open-sky case, where the victim receiver is placed in a playground with no significant surrounding blockage, as shown in Fig. 3(a). It provides a clean reference for evaluating spoofing effects under stable tracking conditions. Scenario 2 is a pedestrian suburban case, where the receiver is carried by a pedestrian moving at approximately 0–2 m/s in a coastal suburban area, as shown in Fig. 3(b). The environment is mostly open, with a building north of the trajectory that reflects signals from the southern sky. At approximately 31 s, PRNs 5 and 22 are identified by ray-tracing analysis as multipath-affected signals. This scenario provides a low-dynamic case with relatively stable and interpretable multipath effects. Scenario 3 shown in Fig. 3(c) is a vehicle urban-driving case, where the receiver moves from a relatively open area into a dense urban environment. Compared with scenarios 1 and 2, it introduces stronger receiver dynamics, more frequent blockage and multipath, and less

stable tracking conditions, making it suitable for evaluating spoofing effects under realistic urban vehicle navigation. Scenario 4 in Fig. 3(d) is a simulation-based reference case, where both the authentic and spoofing signals are generated by the signal simulator. Since no environmental propagation effects are included, it enables more precise synchronization and is mainly used for error characterization, synchronization validation, and baseline comparison under ideal signal-generation conditions.

1. Ray-Tracing-Based Propagation-State Labeling

Environmental propagation effects are labeled in the time domain using ray-tracing. Based on the receiver ground-truth trajectory, satellite ephemeris, and nearby 3D building models from the Hong Kong Lands Department, the ray-tracing algorithm estimates feasible satellite-to-receiver propagation paths. The model assumes specular reflection from building surfaces and checks whether reflected paths geometrically intersect the building surfaces (Fang et al., 2024; Zhang et al., 2021). For each satellite and epoch, the propagation state is classified into four categories: LOS, when only the direct path is available; multipath (MP), when the direct path coexists with one or more reflected paths; NLOS, when the direct path is blocked but at least one reflected path is available; and blocked/unavailable (NaN), when no feasible direct or reflected path is identified. The ray-tracing labels are provided at 1 s intervals and released with the dataset to support the interpretation of authentic signal degradation. Since ray-tracing is based on simplified geometric optics, it cannot fully capture propagation effects caused by irregular structures, vegetation, moving objects, or dynamic occlusion, especially in the more complex urban vehicle environment of scenario 3. The per-satellite propagation-state labels for scenarios 2 and 3 are shown in Fig. 4. For scenarios 1 and 4, all satellites are labeled

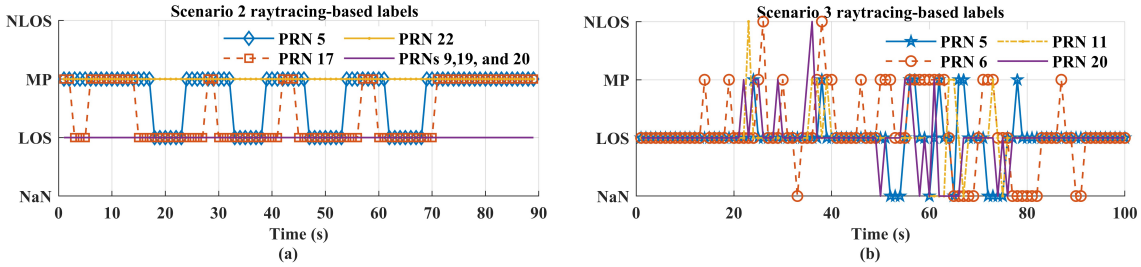


Figure 4: Ray-tracing-based per-satellite propagation-state labels for (a) scenario 2 and (b) scenario 3.

as LOS throughout the entire recording period.

IV. SPOOFING TEST OVERVIEW

Based on the four clean scenarios, multiple spoofing tests are conducted by varying the spoofing signal power advantage and spoofed positioning error. The detailed spoofing settings are summarized in Table 3, and the released files for each scenario are listed in Table 4. For each scenario, an unspoofed clean test is obtained by replaying only the authentic signal under the same laboratory setup, representing the nominal receiver behavior. The receiver ground truth is provided by a NovAtel SPAN-CPT GNSS/INS system, including time-tagged position, velocity, and attitude information. Ray-tracing-based propagation labels are also provided for the dynamic scenarios. The current release contains GPS L1 recordings only, while data from other constellations, such as Galileo and BeiDou, will be progressively added in future updates. The dataset is available at: https://github.com/Fangjingxiaotao/HK-UrbanSpoof.

1. Spoofing Tests in Scenario 1

Scenario 1 provides a static open-sky reference for evaluating spoofing effects under stable authentic tracking conditions. The spoofing signal is injected at approximately 40 s and starts the pull-off process at approximately 60 s. As shown in Fig. 5(a), the spoofed position error gradually increases and reaches approximately 750 m by the end of the test. Two spoofing tests are conducted using the same attack trajectory but different spoofing signal power advantages of approximately 4 dB and 11 dB. The C/N_0 level in the unspoofed data may not be exactly identical to that in the spoofed tests because of slight variations in the RF transmission chain, hardware configuration, and replay conditions between repeated laboratory trials. Fig. 5(b) shows the in-phase prompt correlator output of PRN 16. After spoofing injection, the prompt output generally increases as the spoofing signal power rises and begins to dominate the authentic component. The coherent combination of the authentic and spoofing components depends on their relative code delay, Doppler frequency, and carrier phase. Therefore, close carrier-phase alignment during injection is important for achieving a smooth transition in the correlator output and is also a major cause of failed spoofing trials. Fig. 5(c) shows the corresponding C/N_0 measurements. During injection and early power increase, C/N_0 does not necessarily increase; for example, S1_T2 shows temporary decreases or oscillations due to the interaction between multiple signal components. This effect may persist until the spoofing and authentic components are sufficiently separated in

Table 3: Configuration of the spoofing tests

File ID	Scenario	Injection time (s)	Pull-off start time (s)	File length (s)	Spoofing power advantage (dB)	Target spoofing distance (m)
S1_T1	1: Static; Opensky	41	60	90	4	750
S1_T2					11	750
S2_T1	2: Low Dynamic; Sub-urban	38	50	87	3	600
S2_T2					6	600
S2_T3					7	600
S2_T4					1.5	300
S2_T5					4	300
S2_T6					6	300
S2_T7					3.5	90
S2_T8					7	90
S2_T9					11	90
S2_T10					6	30
S2_T11					10	30
S3_T1	3: High Dynamic; Urban	38	50	95	3.5	600
S3_T2					5	600
S3_T3					6	600
S4_T1	4: Static; Opensky (simulated)	41	60	118	3	400

Table 4: Summary of released data types in each scenario.

Released data type	**Scenario 1**	**Scenario 2**	**Scenario 3**	**Scenario 4**
Clean IF data	S1_Clean	S2_Clean	S3_Clean	S4_Clean
Spoofed IF data	S1_T1, S1_T2	S2_T1-S2_T11	S3_T1-S3_T3	S4_T1
Receiver ground truth	–	S2_GT	S3_GT	–
Ray-tracing-based label	–	S2_Ray	S3_Ray	–

code phase, typically beyond the receiver correlator span. Compared with S1_T2, S1_T1 shows a more subtle C/N_0 variation, making the injection less obvious from simple power monitoring. Fig. 5(d) shows that the Doppler frequency of PRN 16 evolves consistently with the spoofed code-phase trajectory during pull-off.

2. Spoofing Tests in Scenario 2

Scenario 2 evaluates spoofing attacks in a pedestrian suburban environment with low receiver dynamics and relatively stable multipath conditions. The spoofing tests are grouped by the final spoofed positioning error, approximately 600 m, 300 m, 90 m, and 30 m, with different spoofing signal power advantages within each group. Fig. 6 shows representative positioning results from different error levels. The unspoofed solution remains close to the ground truth, indicating nominal receiver performance under authentic GNSS signals. The spoofed trajectory of S2_T6 is directed eastward, while the other selected spoofed trajectories are mainly directed southward. As shown in Fig. 6(b), the spoofing-induced position error becomes more evident after approximately 50 s, corresponding to the pull-off stage. The receiver coordinate errors are computed as

$$\Delta X = X_{\mathrm{GT}} - \hat{X}, \quad \Delta Y = Y_{\mathrm{GT}} - \hat{Y}, \quad \Delta Z = Z_{\mathrm{GT}} - \hat{Z}, \tag{2}$$

where $(\hat{X}, \hat{Y}, \hat{Z})$ denotes the receiver-estimated position and $(X_{\mathrm{GT}}, Y_{\mathrm{GT}}, Z_{\mathrm{GT}})$ denotes the ground-truth reference position.

Receiver measurement performance in the 600 m tests. Tests S2_T1, S2_T2, and S2_T3 use the same spoofed trajectory with a final positioning error of approximately 600 m, while the spoofing signal power advantages are approximately 3 dB, 6 dB,

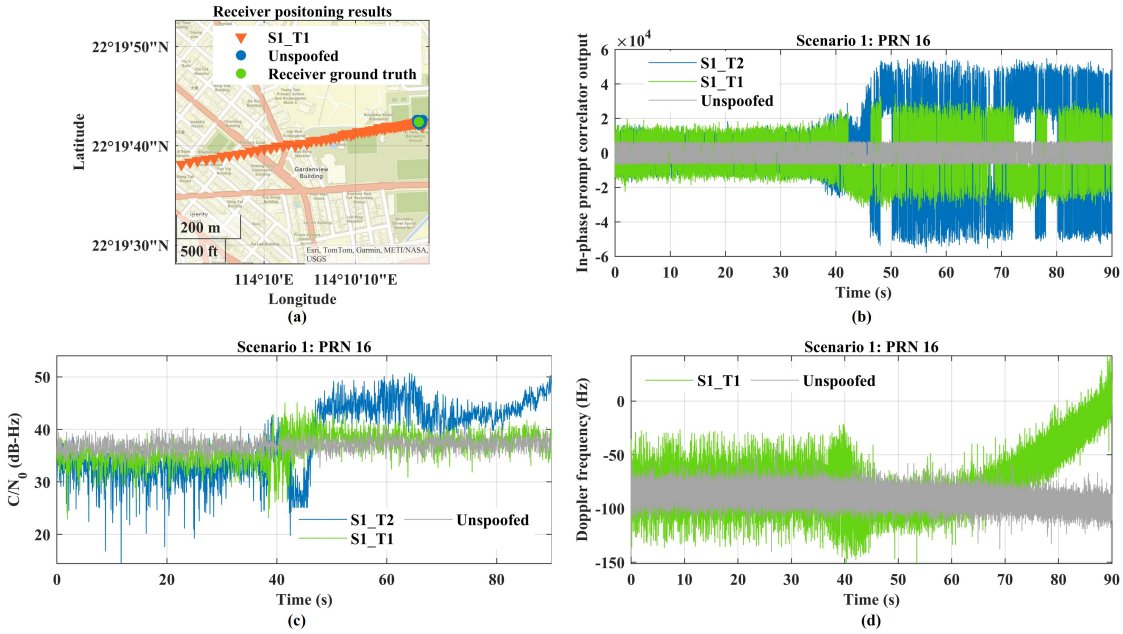


Figure 5: Receiver measurements for the unspoofed test and spoofed tests in scenario 1: (a) receiver position solution and ground truth, (b) in-phase prompt correlator output, (c) C/N_0, and (d) Doppler frequency of PRN 16.

and 7 dB, respectively. Fig. 7(a) shows the in-phase prompt correlator output of PRN 20. The prompt correlator amplitude starts to increase at approximately 38–40 s, corresponding to the spoofing signal injection and power ramp-up stage. The amplitude becomes more stable after approximately 50 s, when the spoofing signal starts to drag the receiver tracking point away from the authentic signal. Fig. 7(b) shows the corresponding C/N_0 measurements. After spoofing injection, C/N_0 exhibits strong fluctuations at around 40 s. These fluctuations are mainly caused by the interaction between the authentic and spoofing components. The C/N_0 measurements gradually return to a relatively normal level after approximately 60 s, when the spoofing component becomes dominant and the two components are more separated in the code phase. This observation indicates that C/N_0 may provide useful evidence during the injection and early pull-off stages, but it may not serve as a reliable feature for continuous spoofing detection throughout the entire attack.

Fig. 8 (a) compares the Doppler frequency estimates of PRN 22 with and without the spoofing attack in S2_T2. In the unspoofed test, the Doppler frequency exhibits periodic upward and downward variations, which are mainly caused by the pedestrian's reciprocating motion. The Doppler pull-off starts at approximately 50 s and then gradually deviates from the clean Doppler trend. Around 59 s, clear Doppler oscillations are observed in the sub-figure. These oscillations are caused by the coexistence and interaction of multiple signal components within the carrier frequency tracking loop, and their pattern is consistent with the theoretical spoofing-induced Doppler response reported in (Fang and Xu, 2025). However, similar Doppler oscillations are also observed around 35 s, before the spoofing signal is injected at approximately 40 s. Therefore, these oscillations should be attributed to unintentional interference, mainly urban multipath. This interpretation is supported by the ray-tracing labels in Fig. 4(a), where PRN 22 is identified as multipath-affected in scenario 2. Similar multipath-induced Doppler oscillations have also been theoretically and experimentally characterized in (Fang et al., 2024; Li et al., 2024). This comparison shows that Doppler frequency features are effective for indicating the presence of multiple signal components and can be exploited for interference-aware detection and mitigation (Fang et al., 2026). Therefore, Doppler oscillation is not a spoofing-specific feature. This ambiguity highlights the need for our dataset that includes both multipath and spoofing conditions within a unified experimental framework. The proposed dataset enables a more realistic evaluation of anti-spoofing and anti-multipath methods, and provides a useful benchmark for assessing whether a method can distinguish spoofing-induced signal errors from multipath-induced distortions.

Receiver measurement performance in the 300 m tests. Fig. 7(c) and (d) shows the in-phase prompt correlator output and C/N_0 measurements of PRN 20 for tests S2_T4, S2_T5, and S2_T6, where the final spoofed positioning error is approximately 300 m. Similar to the 600 m tests, the prompt correlator output increases during the spoofing power ramp-up stage. S2_T5 shows a lower and more unstable C/N_0 level than S2_T4, even though the physical spoofing signal strength is not necessarily lower. This result indicates that a spoofing signal with a larger amplitude does not necessarily lead to a higher receiver-estimated

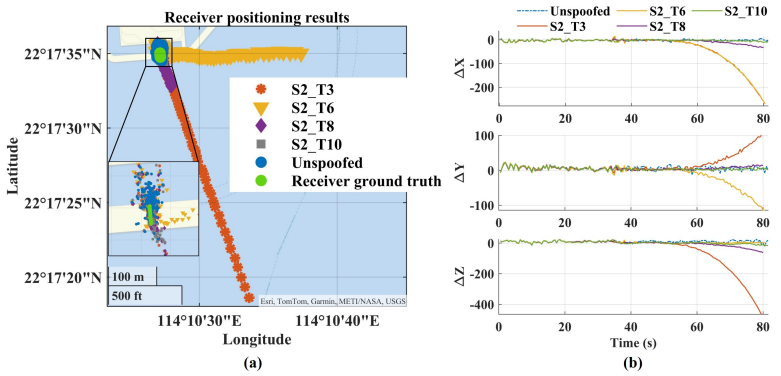


Figure 6: Receiver positioning in scenario 2: (a) receiver positioning results, and (b) receiver coordinate errors ΔX, ΔY, and ΔZ with respect to the ground-truth reference.

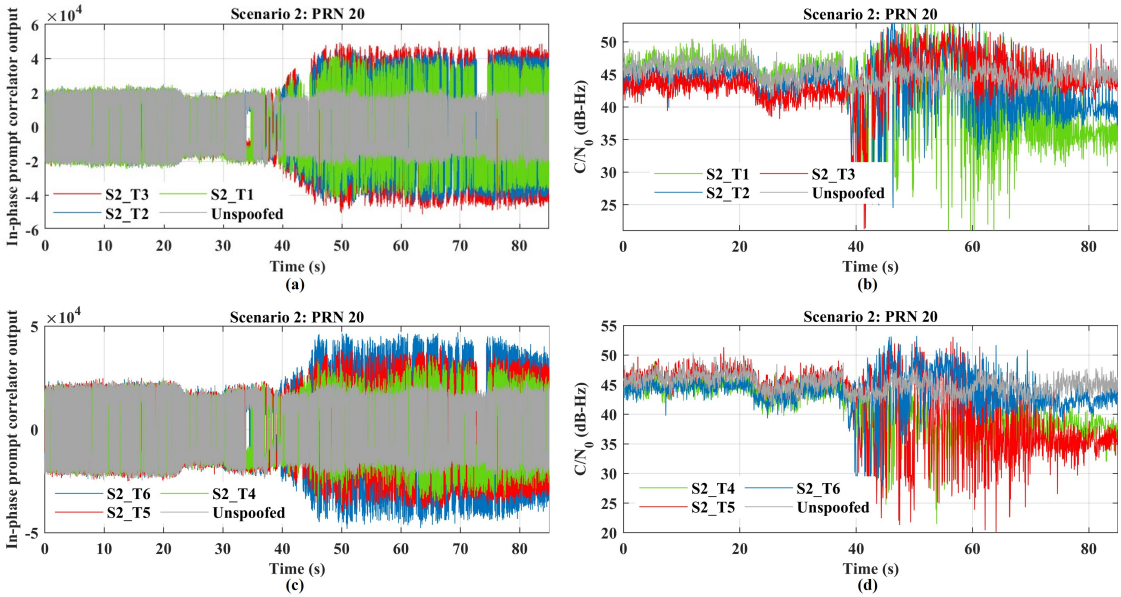


Figure 7: Receiver measurements for PRN 20 in scenario 2: in-phase prompt correlator output and receiver-estimated C/N_0 for S2_T1–S2_T3 in (a)–(b) and S2_T4–S2_T6 in (c)–(d)

C/N_0 if it is not well aligned with the receiver's local replica. The same mechanism also explains the Doppler behavior of S2_T4 and S2_T5 shown in Fig. 8 (b). Compared with S2_T4, test S2_T5 exhibits stronger Doppler oscillations and higher tracking noise. This is likely caused by imperfect carrier-phase alignment and the resulting disturbance in the carrier tracking loop. Such abnormal Doppler fluctuations may increase the probability of spoofing detection.

Receiver measurement performance in the 90 m and 30 m tests. Tests S2_T7, S2_T8, and S2_T9 introduce a final positioning error of approximately 90 m. Their in-phase prompt correlator output and C/N_0 measurements are shown in Fig. 9(a) and (b). Tests S2_T10 and S2_T11 further reduce the final positioning error to approximately 30 m, and the corresponding measurements are shown in Fig. 9(c) and (d). Although the final spoofed positioning error may be small, such attacks can still be critical for applications requiring lane-level or precise positioning accuracy, such as autonomous vehicles and real-time kinematic (RTK)-based navigation. Compared with the 600 m and 300 m tests, these small-error attacks produce a smaller final code-phase separation between the spoofing and authentic signals. As a result, the receiver may track a distorted composite correlation peak rather than a clearly separated spoofing peak. This creates a challenging condition for anti-spoofing methods that require clear separation in the code phase or Doppler frequency before reliable detection or correction. In addition, the spoofer's effect on the receiver measurement can be partially hidden by nominal positioning noise, multipath, and NLOS-induced errors. These tests therefore, provide challenging cases for evaluating whether an anti-spoofing method remains robust in a complex environment.

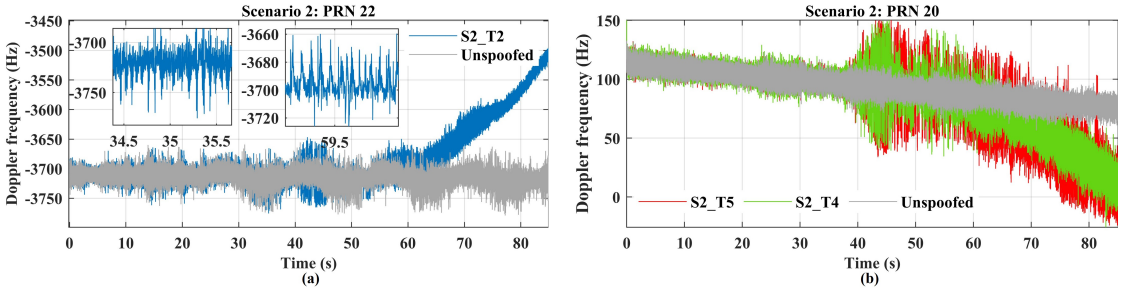


Figure 8: Doppler frequency measurements of (a) PRN 22 and (b) PRN 20 for tests S2_T2 in scenario 2.

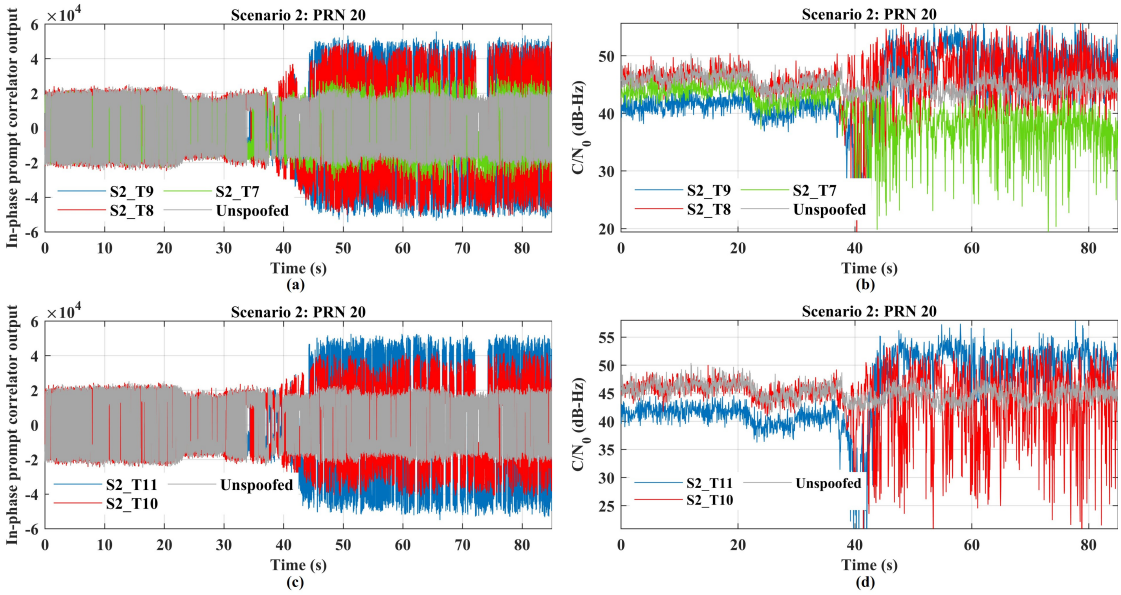


Figure 9: Receiver measurements for PRN 20 in scenario 2: in-phase prompt correlator output and receiver-estimated C/N_0 for S2_T7–S2_T9 in (a)–(b) and S2_T10–S2_T11 in (c)–(d).

3. Spoofing Tests in Scenarios 3 and 4

Spoofing tests in scenarios 3. Scenario 3 represents a vehicle test in a dense urban environment. Fig. 10(a) shows that the spoofing attack gradually drives the receiver solution away from the ground-truth trajectory, leading to a final positioning error of approximately 600 m. The coordinate errors further show that the unspoofed solution contains abrupt error spikes around 55 s, caused by harsh urban propagation conditions. In S3_T1, S3_T2, and S3_T3, the spoofing signal power advantages are approximately 3.5 dB, 5 dB, and 6 dB, respectively. Fig. 11(a) and (b) show the receiver measurement performance. The in-phase prompt correlator output and C/N_0 measurements exhibit noticeable fluctuations caused by signal blockage and urban environmental disturbances. Therefore, although correlator-output distortion and C/N_0 variation provide useful spoofing evidence, they are less distinctive than in cleaner scenarios. Fig. 12(a) shows the Doppler frequency measurements of PRN 11. In S3_T1, the spoofed Doppler fluctuates from around 40 to 50 s, while abrupt disturbances are also observed in the unspoofed case at around 58 s, most likely due to brief tracking instability or loss of lock in the urban canyon. These results show that distinguishing spoofing-induced fluctuations from unintentional urban-induced anomalies may remain challenging.

Spoofing tests in scenarios 4. Scenario 4 represents a fully simulated spoofing attack, where both the authentic and spoofing signals are generated in simulation. Therefore, no environmental interference is involved. In this scenario, the spoofing signal has a power advantage of approximately 3 dB, and the spoofed position is set approximately 400 m away from the authentic receiver position. As shown in Fig. 10 (b), the receiver solution is smoothly pulled away from the ground truth. Fig. 11(c) and (d) show the corresponding receiver measurements. Unlike the real urban scenarios, this simulated case does not show strong C/N_0 oscillations during the injection and pull-off stages. This is because the simulated authentic and spoofing signals are more accurately aligned. The Doppler frequency in Fig. 12 (b) also follows a smooth trend. In particular, no strong Doppler

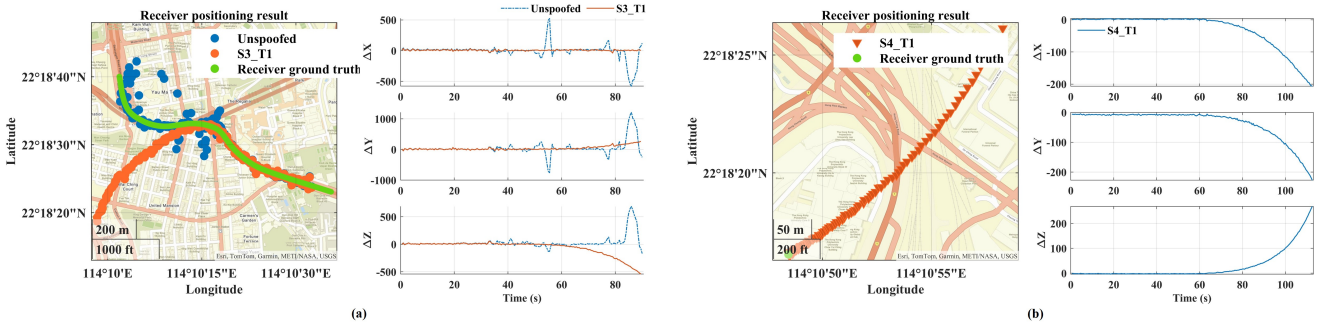


Figure 10: Receiver positioning results in (a) scenario 3 and (b) scenario 4, including the estimated trajectory/position, ground-truth reference, and coordinate errors ΔX, ΔY, and ΔZ.

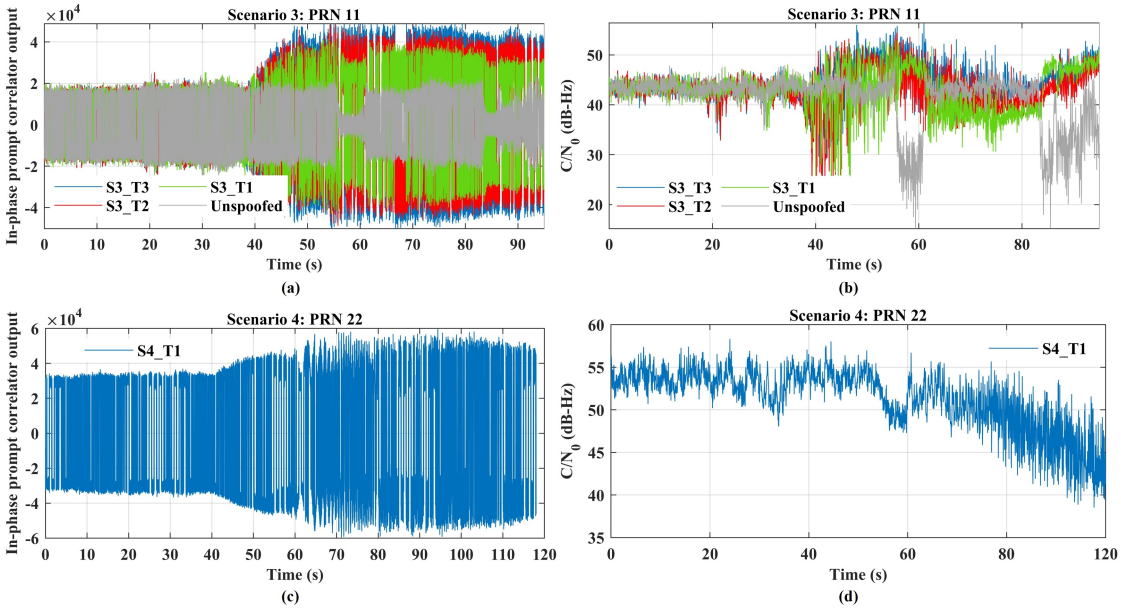


Figure 11: Receiver measurements in scenarios 3 and 4: in-phase prompt correlator output and receiver-estimated C/N_0 for PRN 11 in S3_T1–S3_T3, shown in (a)–(b), and for PRN 22 in S4_T1, shown in (c)–(d).

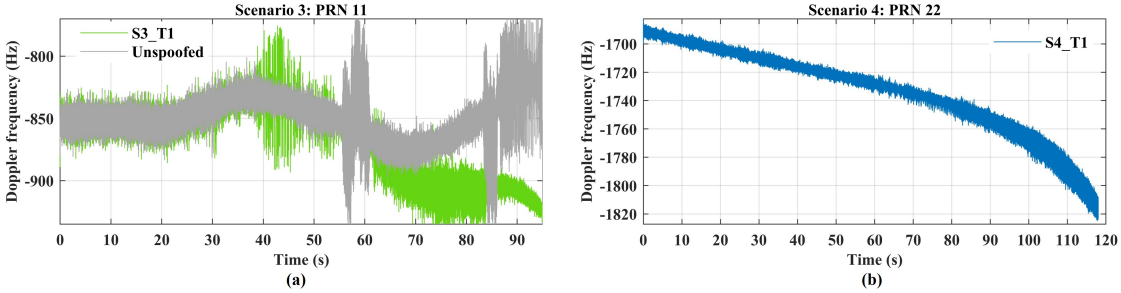


Figure 12: Doppler frequency measurements of (a) PRN 11 for S3_T1 in scenario 3 and (b) PRN 22 for S4_T1 in scenario 4.

oscillation is observed around the beginning of the pull-off stage, in contrast to the fluctuations observed in the real urban tests shown in Figs. 12 (a) and 8. Compared with the real urban scenarios, scenario 4 demonstrates a more ideal and more stealthy spoofing pattern. These results suggest that future spoofing attacks with more accurate signal synchronization may not produce easily detectable abnormalities in the receiver tracking loop and measurements. Therefore, spoofing detection should not rely solely on obvious jumps and excessive measurement noise. Deeper and more continuous features in receiver observables can be modeled and characterized throughout the smooth takeover process.

4. Comparison with TEXBAT and OAKBAT

As summarized in Table 5, the proposed dataset does not contain a test case that is fully identical to those in TEXBAT or OAKBAT. Nevertheless, two groups of comparable scenarios can be identified. The first group, including S1_T2, TEXBAT_ds2, and OAKBAT_os2, corresponds to static-user cases with a relatively high spoofing-to-authentic power advantage. The second group, including S3_T1, TEXBAT_ds6, and OAKBAT_os6, represents dynamic-user cases with moderate or small spoofing power advantages. A key difference lies in the code-carrier phase relationship between the spoofing and authentic signals. Several TEXBAT and OAKBAT scenarios are generated under a frequency-locked condition, where the spoofing signal is intentionally controlled to maintain a stable carrier frequency relationship with the authentic signal during takeover. This setting can reduce carrier tracking loop oscillations and enables successful spoofing with a very small power advantage, for example, 0.8 dB in TEXBAT_ds6 and OAKBAT_os6. However, maintaining frequency lock while shifting the code phase may also introduce code–carrier inconsistency, which can be exploited by code-minus-carrier (CMC)-based detection methods (Blum et al., 2021; Feng et al., 2024). In contrast, the proposed dataset is generated under the frequency-unlocked mode. As a result, a larger spoofing power advantage is required to achieve a successful takeover. In our tests, at least 1.5 dB of spoofing power advantage was needed. Fig. 13 compares the multi-correlator outputs of three representative tests. For each

Table 5: Comparison of spoofing scenarios in different datasets

File ID	Spoofing type	User mobility	Power advantage (dB)	Frequency lock status
S1_T2 (this work)	Position	Static	10–11	Unlocked
TEXBAT_ds2	Time			Unlocked
OAKBAT_os2	Time			Locked
S3_T1 (this work)	Position	Dynamic	3.5	Unlocked
TEXBAT_ds6	Time		0.8	Locked
OAKBAT_os6	Time		0.8	Locked

case, the first row presents the 3D evolution of the correlation function over code phase and time, while the second row provides the corresponding top-view heatmap. In the proposed S3_T1 dataset, the spoofing peak gradually increases and eventually dominates the correlation function. A residual branch of the authentic signal can still be observed toward the end of the test. A similarly smooth code-phase pull-off can also be observed in TEXBAT_ds6. Since OAKBAT is generated using simulated signals, it presents a more idealized correlation pattern, with relatively stable and clearly separated correlation branches. These differences indicate that the proposed dataset provides a realistic and challenging benchmark for evaluating spoofing detection and mitigation methods.

V. CONCLUSION

GNSS spoofing poses a serious threat to satellite-based navigation systems, particularly when a smooth takeover strategy is adopted. In this type of attack, a sophisticated spoofer gradually introduces counterfeit signals while preserving the continuity of the receiver code and carrier tracking loops, making the attack difficult to identify from abrupt measurements. Although substantial progress has been made in GNSS spoofing detection and mitigation, the validation of existing methods still largely relies on a limited number of public datasets or simplified simulation settings. These datasets provide important references for anti-spoofing research, but they do not fully capture the degraded signal reception conditions commonly encountered in dense urban environments.

To address this gap, this paper presents an urban GNSS spoofing dataset (HK-UrbanSpoof) designed to support more comprehensive evaluation of anti-spoofing and anti-multipath methods. The dataset covers four experimental environments, ranging from open-sky and coastal suburban areas to dense urban scenarios. It includes 17 spoofing tests with different spoofing signal power advantages and false trajectories, covering static, pedestrian, and vehicle cases. To facilitate the interpretation of authentic signal degradation, ray-tracing analysis is provided to label the propagation conditions of authentic GNSS signals. In addition, the released receiver ground truth includes position, velocity, and attitude information obtained from a GNSS/INS

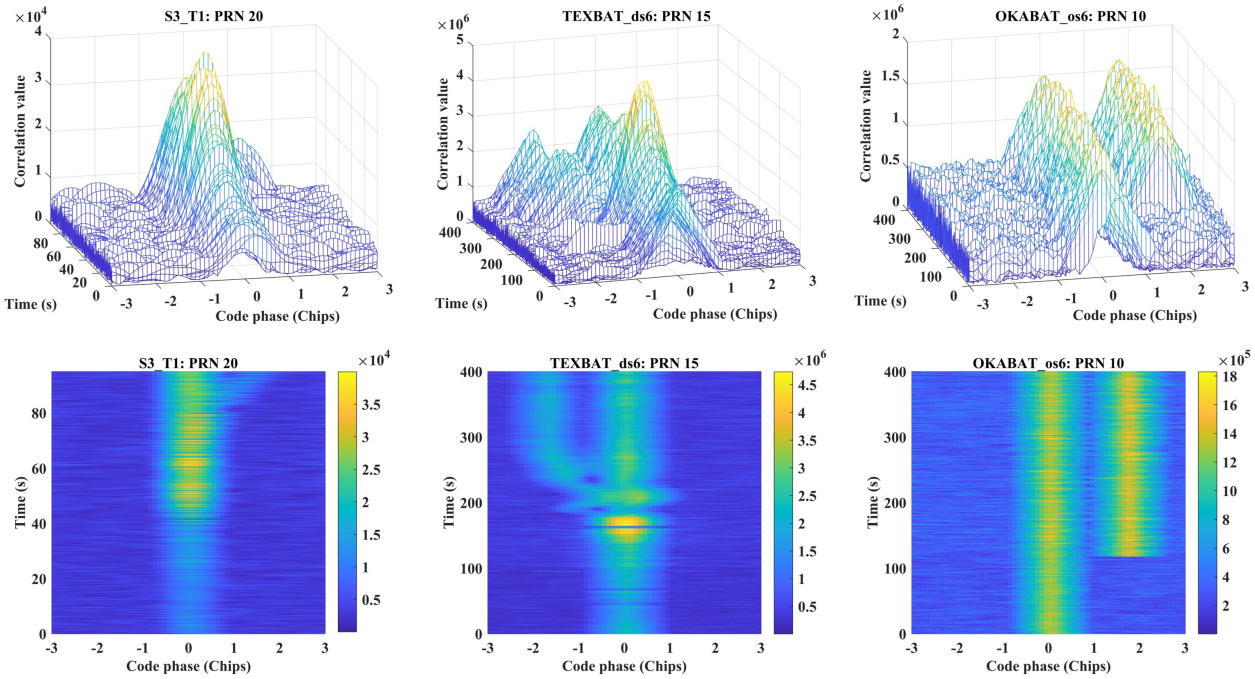


Figure 13: Multi-correlator outputs of: S3_T1, PRN 20; TEXBAT_ds6, PRN 15; and OAKBAT_os6, PRN 10.

system, enabling navigation-level evaluation and further position-based research. This work also documents the authentic GNSS signal reception platform, spoofing signal generation platform, hardware implementation workflow, and receiver measurement behavior associated with the dataset. The experimental results show that the spoofer can seamlessly shift the receiver position without causing obvious jumps or excessive measurement noise. In particular, the dataset includes small-position-error spoofing attacks, where the spoofer attempts to mimic realistic urban positioning errors rather than generating large and easily detectable positioning offsets. A simulated spoofing scenario is also released to support error characterization, synchronization validation, and baseline comparison under ideal signal-generation conditions.

Overall, the proposed dataset provides challenging and realistic spoofing scenarios that may evade conventional anti-spoofing methods. It offers a useful benchmark for evaluating whether a method can distinguish spoofing-induced signal distortions from environment-induced effects. The dataset therefore, contributes to the development and validation of more robust, generalizable, and practically deployable GNSS anti-spoofing and interference-mitigation techniques.

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